

MIANEH, Iran, Islamic Republic of

WMO: 407160

Lat: 37.450N

Lon: 47.700E

Elev: 1110

StdP: 88.68

Time Zone: 3.50 (IRN)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.1	(c) -8.3	(d) -15.1	(e) 1.1	(f) -8.9	(g) -12.3	(h) 1.5	(i) -5.9	(j) 9.3	(k) 10.3	(l) 8.2	(m) 7.5	(n) 0.5	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.2	(c) 38.8	(d) 18.5	(e) 37.2	(f) 18.1	(g) 35.7	(h) 18.0	(i) 20.3	(j) 33.7	(k) 19.6	(l) 33.0	(m) 19.0	(n) 32.6	(o) 4.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.9	26.0	14.9	12.1	26.1	14.1	11.5	25.3	63.7	33.9	61.0	32.7	58.9	32.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.3	(b) 8.6	(c) 7.5	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(4) 10.3	(5) 8.6	(6) 7.5	(7) WB	(8) -14.3	(9) 22.0	(10) 4.2	(11) 1.5	(12) -17.4	(13) 23.0	(14) -19.8	(15) 23.9	(16) -22.2	(17) 24.7	(18) -25.3	(19) 25.8	(20) 25.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	0.1	2.9	9.1	14.0	19.7	25.6	28.7	28.0	23.3	15.9	7.9	2.5
	DBStd	10.51	4.87	4.69	4.24	3.64	2.86	2.69	2.63	2.42	2.72	3.77	3.82	3.75
	HDD10.0	899	306	199	66	9	0	0	0	0	0	3	83	233
	HDD18.3	2337	564	432	286	138	18	0	0	0	2	93	314	491
	CDD10.0	2681	0	1	39	127	302	469	579	558	400	187	18	1
	CDD18.3	1078	0	0	1	6	62	219	321	299	152	18	0	0
	CDH23.3	12676	0	0	14	94	650	2468	4006	3657	1582	205	0	0
	CDH26.7	6682	0	0	1	21	200	1279	2347	2091	701	42	0	0
Wind	WSAvg	2.5	1.3	1.5	2.6	2.6	2.9	3.8	4.0	3.9	3.1	2.3	1.1	1.4
Precipitation	PrecAvg	289	30	26	36	49	42	10	6	3	6	20	33	28
	PrecMax	421	78	60	91	96	74	28	32	20	39	74	112	79
	PrecMin	182	5	4	9	7	15	1	0	0	0	0	1	1
	PrecStd	61	16	13	21	24	18	8	8	5	8	19	28	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	16.1	25.3	29.2	32.7	38.9	41.0	40.1	36.1	30.1	20.2	14.6
		MCWB	6.1	7.9	10.9	13.2	15.6	18.8	18.8	18.2	17.5	14.8	10.6	7.3
	2%	DB	10.6	13.6	21.7	25.9	30.8	36.4	39.5	38.2	34.2	27.9	17.9	11.8
		MCWB	5.2	7.0	10.4	13.0	15.1	17.6	18.8	17.9	17.0	14.1	10.1	6.2
	5%	DB	8.7	11.5	19.1	23.5	29.0	34.6	37.8	37.0	32.6	25.8	15.9	9.7
		MCWB	4.3	6.0	9.1	12.4	14.6	17.3	18.4	18.1	16.7	13.5	9.3	5.4
	10%	DB	6.6	9.6	16.8	21.3	27.1	33.0	36.0	35.5	30.9	23.6	14.1	7.9
		MCWB	3.3	5.2	8.4	11.4	14.3	16.7	18.4	18.0	16.3	12.9	8.8	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	9.4	12.2	14.9	19.3	21.9	21.2	20.9	19.5	16.7	12.7	8.2
		MCDB	11.1	14.0	22.1	22.5	26.5	34.3	36.0	34.1	31.4	26.5	16.7	12.6
	2%	WB	5.7	7.7	10.9	13.9	16.8	19.6	20.3	20.1	18.6	15.2	11.3	6.9
		MCDB	9.2	11.9	20.1	22.4	25.4	32.7	34.7	33.0	30.8	24.3	15.2	10.0
	5%	WB	4.6	6.6	9.7	13.1	15.8	18.5	19.7	19.4	17.8	14.3	10.2	5.9
		MCDB	8.0	10.5	17.7	21.4	25.4	31.7	33.6	32.5	29.7	22.8	14.4	8.9
	10%	WB	3.6	5.6	8.8	12.2	15.1	17.5	19.1	18.8	17.0	13.5	9.2	5.0
		MCDB	6.6	9.1	15.8	19.8	24.9	30.5	33.2	32.1	28.5	21.5	13.6	7.6
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.7	11.7	12.5	13.1	14.4	14.2	14.6	14.7	12.5	10.3	8.9
		MCDBR	11.5	12.8	15.5	15.8	15.3	15.7	15.6	16.0	16.5	16.2	13.6	11.3
		MCWBR	6.9	7.2	7.0	6.7	5.0	5.2	4.8	5.3	6.1	6.7	7.1	6.8
	5% WB	MCDBR	9.9	10.9	13.5	13.3	13.5	14.6	14.2	14.6	14.7	13.8	11.1	9.6
		MCWBR	6.2	6.2	6.3	6.1	5.2	5.4	4.6	4.9	5.4	6.0	5.8	5.9
Clear-Sky Solar Irradiance	taub	0.327	0.366	0.427	0.508	0.503	0.482	0.524	0.482	0.423	0.417	0.350	0.321	
	taud	2.276	2.160	2.007	1.843	1.894	1.969	1.887	1.985	2.121	2.080	2.255	2.316	
	Ebn at Noon	844	850	827	780	793	808	770	794	822	778	806	825	
	Edh at Noon	99	126	162	203	196	182	196	174	143	135	100	89	
All-Sky Solar Radiation	RadAvg	2.51	3.43	4.29	5.22	6.55	7.87	7.71	7.09	5.79	3.97	2.66	2.12	
	RadStd	0.21	0.36	0.41	0.47	0.36	0.41	0.20	0.19	0.26	0.30	0.35	0.29	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.33	N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page